

**KNEE OSTEOARTHRITIS PREDICTION AND
PROGRESSION USING MULTI-MODAL DEEP
LEARNING.**

Project ID: 25-26J-112

BSc (Hons) in Information Technology Specializing in Data Science

Department of Information Technology
Faculty of Computing

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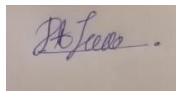
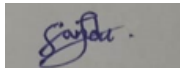
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ABSTRACT

The diagnosis, grading and monitoring of Knee Osteoarthritis are essential in today's healthcare, but have several challenges, largely because of the limited effectiveness of single-modality assessment and absence of real-time monitoring from the patient's perspective. This study introduces a multi-modal approach to early detection, risk prediction and monitoring of KOA, comprising four components: prediction from imaging, clinical and biomarker-based assessment, IoT-based continuous monitoring and intelligent severity grading with treatment recommendation. The first component involves deep learning-based models for X-ray and MRI image analysis, which automatically detects KOA by assessing bone deformations and soft tissue changes. The second component deals with structured clinical, demographic, and biomarker information, in which machine learning techniques are used to predict the presence and risk of the disease, helping to improve early diagnosis in resource-poor scenarios. The third component is a Smart IoT-based wearable monitoring system that incorporates VAG signals and biomechanics data, with embedded TinyML models on ESP32 microcontrollers to perform real-time, low-power, and continuous monitoring of knee health beyond clinical settings. The fourth component of the system uses a fusion-based approach to integrate these diverse data streams for accurate severity grading according to the Kellgren-Lawrence scale, and a rule-based system to offer treatment and follow-up advice.

These components showcase the complementary nature of deep learning, machine learning, IoT, and explainable AI approaches in designing a holistic and patient-friendly KOA monitoring system. This approach not only enhances the diagnostic accuracy and explainability but also facilitates monitoring and intervention. The results demonstrate the benefits of multi-modal fusion in bridging the gap between diagnosis and treatment and enable scalable, cost-effective, and intelligent monitoring solutions for degenerative joint conditions.

Keywords: Knee Osteoarthritis, Vibroarthrography, X-ray, MRI, Clinical Data, IoT, Multi-Modal Learning, Deep Learning, Explainable AI, Severity Grading

TABLE OF CONTENTS

DECLARATION.....	iii
LIST OF FIGURES	viii
LIST OF ABBREVIATIONS	ix
1. INTRODUCTION.....	1
1.1 Background	1
1.2 Literature Review	4
1.3 Research Gap	6
1.4 Research Problem	9
1.5 Research Objectives.....	12
1.5.1. Main Objective	12
1.5.2. Sub-Objectives.....	12
2. METHODOLOGY	15
2.1 Methodology	15
2.1.1. System Overview	15
2.1.2. Data Collection and Preprocessing.....	17
2.1.3. Functional, Non-Functional Requirements	19
2.2 Commercialization Aspect of the Product	24
2.2.1. Market Potential	25
2.2.2. Business Model and the Revenue Stream	25
2.2.3. Scalability & Adaption	25
2.3 Testing and Implementation	26
2.3.1. Implementation	26
2.3.2. Testing.....	29
3. RESULTS	33
3.1 Results	33
3.2 Research Findings.....	39
3.3. Discussion.....	42
4 SUMMARY OF INDIVIDUAL CONTRIBUTION	46
4.1. JAYASINGHE J M N S (IT22582942).....	46
4.2. FERNANDO W D A (IT22223708)	46
4.3. PERERA B B A R (IT22606792)	46
4.4. GAMAGE D.M.G.P.K (IT22188472)	47
5 CONCLUSION	48
REFERENCES.....	50

GLOSSARY.....	52
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LIST OF FIGURES

Figure 1- Normal Knee Vs Osteoarthritis Knee.....	1
Figure 2 - Kellgren Lawrence scale	2
Figure 3 - Confusion Matrix of the Imaging Component (X-Ray).....	33
Figure 4 - Results of the X-Ray Prediction Module	34
Figure 5 - Confusion Matrix of the Imaging Component (MRI).....	34
Figure 6 - Results of the MRI Prediction Module.....	34
Figure 7 - Learning Curve of the Final Clinical Model	35
Figure 8 - Confusion Matrix of the Final Clinical Model.....	36
Figure 9 - EfficientNetB0 Model Confusion matrix	37
Figure 10 - EfficientNetB0 Model Accuracy Curve	37
Figure 11 - Confusion matrix of the XGBoost clinical model.....	38
Figure 12 - Confusion Matrix of the Knee Monitoring Module	38

LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
B2B	Business to Business
B2C	Business to Consumer
BMI	Body Mass Index
CNN	Convolutional Neural Network
COMP	Cartilage Oligomeric Matrix Protein
CRP	C- Reactive Protein
CT	Computed Tomography
CTX	Cerebrotendinous Xanthomatosis
EHR	Electronic Health Record
EEMD	Ensemble Empirical Mode Decomposition
EMD	Empirical Mode Decomposition
Grad-CAM	Gradient-weighted Class Activation Mapping
IoT	Internet of Things
KL	Kellgren Lawrence
KOA	Knee Osteoarthritis
LIME	Local Interpretable Model-agnostic Explanations
ML	Machine Learning
MRI	Magnetic Resonance Imaging
RMS	Root Mean Square
RoI	Region of Interest
SHAP	Shapley Additive exPlanations
SMOTE	Synthetic Minority Over-sampling Technique
VAG	Vibroarthrography

1. INTRODUCTION

1.1 Background

The emergence of digital health and artificial intelligence technologies has revolutionized diagnostic and disease management practices from discrete clinical evaluations to continuous, data-driven, and patient-focused healthcare services. In this new era, the treatment of chronic musculoskeletal diseases, such as KOA, requires more than traditional diagnostic methods. KOA, a debilitating degenerative joint disease involving cartilage degradation, pain, stiffness and limited functionality, is increasingly becoming a significant global health issue, especially in the elderly. With the growing incidence of KOA, there is a growing demand for smart systems that can not only identify the condition early in its progression but can track its evolution and aid diagnosis and decision-making in a comprehensive and user-friendly way. The future of health care systems will not rely only on the diagnostic precision achieved in a clinical environment but must develop into a dynamic and integrated system that continuously processes a range of patient data and delivers real-time insights to enhance patient outcomes and quality of life.

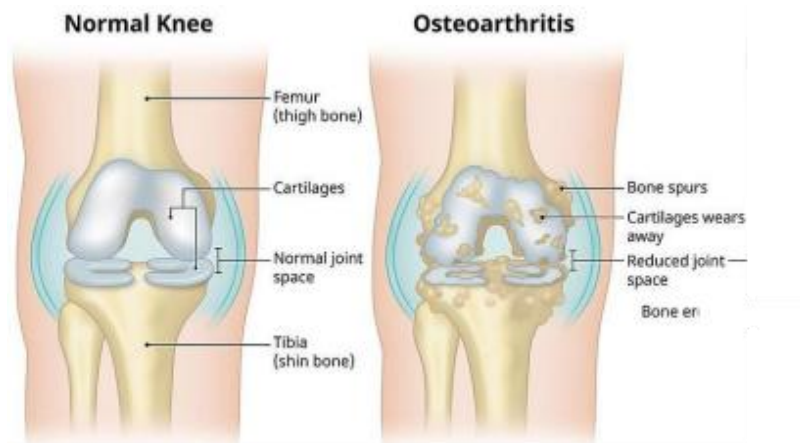


Figure 1- Normal Knee Vs Osteoarthritis Knee

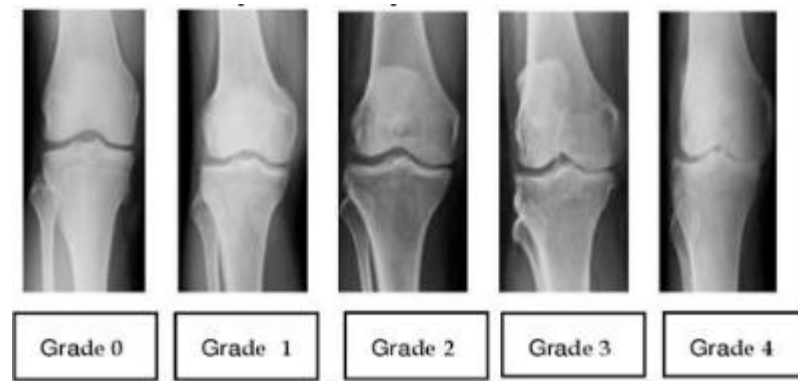


Figure 2 - Kellgren Lawrence scale

Conventional KOA diagnostic techniques largely depend on imaging technologies such as X-rays and MRI, which offer important anatomical insights into the knee joint. X-rays are commonly used to diagnose bone-related pathologies and to assess the severity of the disease using the Kellgren-Lawrence grading system, whereas MRI provides a more detailed view of soft tissues including cartilage, ligaments and synovial structures. While these methods are clinically successful, they are inherently limited by their sporadic nature, expense, and reliance on medical institutions. And diagnosis using imaging is generally performed at a later stage of disease, limiting the potential for early detection. These gaps point to the need for more convenient and preventative diagnostic strategies that are not limited to hospitals and allow for ongoing monitoring of knee joint status.

Meanwhile, the growing availability of structured clinical, demographic, and biomarker data has provided opportunities for early prediction of disease using machine learning approaches. Age, BMI, medical history, symptoms, and laboratory variables offer valuable information on the risk factors and biological mechanisms contributing to KOA. Machine learning can adeptly learn complex, non-linear patterns from this data, allowing early prediction of disease status, even prior to imaging evidence of structural change. But current clinical-based systems are not integrated with imaging systems, resulting in an inability to offer a reliable diagnosis. Moreover, the interpretability of predictive models is often lacking, making it difficult for clinicians to adopt such approaches - they must be able to understand and interpret the results.

Both imaging and clinical data provide essential insights, but they are based on snapshots of a dynamic and evolving disease. A major gap in current KOA studies is

the lack of real-time monitoring techniques to assess functional changes in joint dynamics during everyday activities. Recent advances in IoT technologies and wearable sensors have provided opportunities for continuous monitoring. Methods such as VAG, which capture and process joint vibration signals during dynamic activities, provide a non-invasive way to identify changes in joint biomechanics indicative of cartilage damage. Coupled with embedded computing and low-power machine learning (e.g. TinyML on microcontrollers), these wearable technologies provide real-time analysis on the device, eliminating the need for continuous cloud connectivity. While promising, current implementations are mostly confined to standalone analysis and lack integration with other clinical and image data.

In addition to monitoring and detection, successful management of KOA also depends on severity grading and treatment strategies. Severity assessment, which is usually linked to the Kellgren-Lawrence scale, is a vital part of the decision-making process to decide on appropriate treatment, from lifestyle advice and physical therapy to surgery in severe cases. But, current severity classification approaches are heavily reliant on image analysis, often ignoring other clinical signs and patient-specific characteristics. And treatment suggestions are typically not linked to automated predictive models, leading to a disconnect between diagnosis and clinical recommendations.

These issues collectively point to the need for a gap in the current body of research: the development of an integrated, multimodal system that can combine imaging, clinical and biomarker information with continuous sensor-based monitoring into an intelligent system. To bridge this gap, we need to move beyond single-modality-based approaches and develop an integrated system that can merge multimodal data, improve prediction accuracy, and enable ongoing patient monitoring. This system should not only provide accurate predictions but also be interpretable, scalable and feasible in clinical practice.

This work seeks to overcome these challenges by developing a multi-modal system for predicting and grading KOA, and monitoring the disease. The system integrates four main components: a deep learning-based system for X-ray and MRI image processing to automatically detect disease, a machine learning-based system for clinical and biomarker data analysis to predict the risk of developing KOA, an IoT-

enabled wearable monitoring system based on VAG signals to assess knee health in real-time, and a fusion-based decision support system that uses these components to accurately grade KOA severity and provide personalized treatment recommendations. The use of explainable artificial intelligence ensures transparency and trust in the eyes of healthcare practitioners.

These elements together constitute an integrated, dynamic healthcare system that shifts KOA care from a reactive, hospital-centered approach to a proactive, patient-centered one. The system's ability to perceive, interpret and act based on patient data from physical, clinical and functional perspectives, through deep learning, machine learning, IoT and explainable AI, allows for monitoring and management of patient conditions in real-time. This multi-modal framework not only enhances disease detection and early prediction but also connects clinical assessment and patient monitoring in the real world. In essence, this work paves the way for future smart health systems, in which multi-modal data integration and adaptive decision-making transform how chronic conditions like KOA are diagnosed, assessed and treated.

1.2 Literature Review

Knee Osteoarthritis is a disease characterized by gradual joint destruction requiring early detection, precise evaluation of the degree of the disease, and constant monitoring for reducing disability. It is proven that currently AI is very effective in KOA diagnosis and predicting the outcomes of its treatment due to its ability to detect hidden patterns in medical images, clinical data, and other information about patients [1]. However, KOA is considered a multifactorial disease, and thus there is no way to capture its course through a single source of information.

Diagnostic imaging is one of the most explored subjects of KOA studies. This type of diagnosis is usually performed using plain radiographs, which are inexpensive and accessible in clinical practice. Recently, deep learning algorithms were created that allow for automatic KOA detection from X-ray images, minimizing the need for radiologists' involvement [5]. Moreover, researchers have implemented CNN to determine the degree of KOA severity according to radiograph characteristics like narrowing of joint space, presence of osteophytes and bone sclerosis [6]. Additionally, semi-supervised learning has been proposed to enhance severity determination for cases with insufficient numbers of labeled medical images [13], [14]. Nevertheless, all

X-ray-based solutions identify only bone abnormalities and cannot indicate early cartilage degradation.

An additional research approach is related to MRI-based techniques since this type of imaging allows for visualizing soft tissues such as cartilage, ligament, and meniscus. The use of machine learning algorithms to predict KOA progression based on MRI data was mentioned in previous works, thus confirming the importance of early detection of structural changes [7]. In recent studies, researchers proposed to apply temporal learning and graph learning models on longitudinal MRI scans to predict KOA progression over time [15]. Although the application of these techniques is promising, MRI scans are rather expensive and not widely available in some medical organizations.

The use of clinical, demographic, and blood biomarker data may be relevant to KOA prediction since such data is easy to obtain for patients. Some factors like age, body mass index, sex, physical activity, previous injuries, and clinical signs were identified as predictive of knee osteoarthritis [3]. Furthermore, studies revealed that specific blood biomarkers and BMI values could be considered predictive of knee osteoarthritis, which confirms the importance of metabolic and inflammatory processes in the development of the disease [2]. It was also noted that machine learning models can be helpful to identify factors of clinical and biochemical importance in predicting the disease [4]. Some comparative analyses of KOA prediction models revealed the potential applicability of models like Random Forest, Gradient Boosting, and XGBoost [11],[12]. However, clinical-data-only models may lack structural evidence from imaging, which can reduce diagnostic confidence.

The idea of using multi-modal learning was developed to solve the problem that single-modality techniques cannot address alone. For example, Panfilov et al. suggested a multi-modal learning method based on transformers for end-to-end prediction of KOA [9]. In addition, Chen et al. developed a unified framework for predicting and analyzing multiple views of longitudinal multi-modal data with missing values since it is common practice that healthcare datasets often contain incomplete information [10]. Both studies prove that multi-modal learning has high predictive power. However, there are plenty of works on multi-modal learning techniques that consider only predictions but neglect real-time monitoring and treatment.

Wearable sensors and IoT technologies present one of the promising directions in KOA management. Biomechanical parameters for knees' movement were measured by means of wearable sensor technology [16]. TinyML technology allows using wearable devices for screening arthritic conditions through the analysis of movement and applying machine learning techniques on the device [17]. The use of mobile health platform and wearable systems for remote monitoring following knee-related operations proved the efficiency of mobile apps for displaying the outcome and alerting patients [18]. IoT systems for rehabilitation purposes also show how affordable real-time monitoring could be achieved for KOA [19]. Still, most of such solutions are developed independently from imaging and predictions models.

Another important issue in AI-based medical systems is explainability. Deep learning models can achieve high accuracy but the black-box nature of these models can reduce clinical trust [8]. In the medical diagnosis domain, clinicians want to know why a model made a prediction. Hence, explainable AI techniques like Grad-CAM, LIME, and SHAP are crucial for boosting transparency and supporting clinical decision-making.

Overall, the present studies demonstrate promising advances in the diagnosis of KOA using X-ray, MRI, clinical data, biomarkers, wearable sensors and multi-modal learning. But most studies examine one piece of the problem. Imaging models help in diagnosis of structure, clinical models help in early risk prediction, wearable systems help in monitoring, and multi-modal models help in improving the prediction accuracy. There is still a need for one integrated framework that combines these components to support KOA detection, severity grading, continuous monitoring, explainability, and treatment recommendation.

1.3 Research Gap

KOA, despite substantial progress in AI, machine learning, medical imaging, and wearable technologies, there are various vital gaps left toward a comprehensive, real-world clinical application solution. In imaging-based diagnosis, clinical prediction, biomarker analysis, and wearable monitoring, existing research presents promising results. However, these efforts are disjointed, not well-integrated, and do not cover all the stages and aspects of effective KOA management, from early detection to progression tracking, grading severity, and aiding treatment.

One of the major gaps refers to imaging-based KOA diagnosis. Deep learning approaches using X-ray images have shown high accuracy in detecting KOA and classifying severity levels using radiographic features such as joint space narrowing and osteophyte formation. Semi-supervised learning approaches have achieved even better results under limited labeled data scenarios. X-ray imaging, however, is limited to identifying structural changes in bones and cannot detect early cartilage degeneration, a key feature for early KOA intervention. MRI-based studies tackle this limitation when providing detailed soft tissue information and progression prediction. However, MRI is costly, inaccessible, and unsuitable for regular or ongoing observation in underfunded settings. Therefore, sole dependence on imaging modalities results in incomplete and belated diagnosis, necessitating the development of complementary approaches.

There is also a critical gap in clinical, demographic, and biomarker-based prediction models. In clinical, demographic, and biomarker-anchored prediction models, there is another important gap; studies have shown that variables such as age, BMI, symptoms, and blood biomarkers play a significant role in KOA prediction [2],[3],[4]. Different machine learning models for risk factor identification and predicting the presence of disease have been successfully applied to such structured datasets [11], [12]. These approaches are complementary to direct structural evidence to joint damage, and are often created independently of the imaging system. This limits the diagnosis reliability, limiting clinical confidence in the inferences. There are often missing values, imbalance between classes, and population variability in clinical datasets, which could pose generalization and performance issues for models.

One major gap in the existing KOA studies is that they do not have continuous monitoring and evaluation in real-time. KOA progresses depending on the physical activity conducted per day, biomechanical stimuli, and movement patterns in the joints. However, most existing studies rely on static data collected at one point in time or during clinical visits; A promising solution that has been proposed in sensing biomechanical data such as gait patterns and joint vibrations in real-time is wearable sensor technologies [16]. Similarly, TinyML-based wearable systems demonstrated early arthritis detection capabilities through on-device analysis [17]. IoT-based platforms and mobile health applications have also been used for rehabilitation and

remote monitoring [18], [19]. These wearable-based systems remain isolated and unintegrated with imaging or clinical prediction models. Therefore, they do not provide holistic insights into the progression of KOA from structural, clinical, and functional perspectives.

Multi-modal learning approaches have been proposed to defeat the limits of a single-modality learning system. Multi-modal learning with transformer-based architectures and unified models have been successful in demonstrating that utilizing multiple data sources can increase prediction success rate and robustness [9], [10]. The results, however, highlight the significance of multimodal architecture in further advancing the fusion of imaging, clinical, and longitudinal features. However, most of these move remain on predictive performance and are often limited to experimental or research use. Other real-world challenges (e.g., missing data, real-time processing, clinical deployment, and system scalability) are not fully addressed. For instance, multi-modal systems often do not utilize continuous monitoring data from wearable devices, which is necessary for monitoring disease progression over time.

Another critical gap is a lack of explainability and transparency in AI-based KOA systems. On the other hand, the deep learning models used to achieve high accuracy are black box systems whose decision-making processes cannot be interpreted by clinicians [8]. Interpretability is necessary for trust and adoptability of healthcare applications. Although some studies have proposed incorporation of explainable AI based techniques, such cases remain rare in complete diagnostic systems. In the absence of adequate explanations, clinicians may be reluctant to place faith in the automated predictions, thus limiting the real-world utility of these technologies.

In addition, there are no decision supporting systems that are integrated to link diagnosis to clinical actions. Most studies end at the prediction or classification stage and provide no clinical actions based on their outputs. Most severity grading based on Kellgren–Lawrence grades is done distinctly using images, usually without involving any clinical indicators or factors. A lack of decision support based on clinical outcomes is also evident, with most studies focusing on prediction or classification and not providing treatment planning or patient management based on their results. This limits their potential for real-world clinical decision making.

Another practical limitation is a lack of scalable, accessible and cost-effective solutions. High computational resource requirements of advanced AI models make deployment in low-resource healthcare settings untenable. Similarly, dependence on costly imaging procedures like MRI limits accessibility for much of the population. Systems combining high-performance prediction and efficient, inexpensive, and deployable technologies (e.g., IoT and embedded machine learning models) are needed.

Hence, the main research gap involves the lack of an integrated, multimodal, and explainable real-time KOA management framework that encompasses imaging (X-ray and MRI), clinical and biomarker data, continuous monitoring via wearable sensors, severity grading, and treatment recommendation within a single system. This is despite relevant individual components being developed in existing studies and presented in isolation. This research fills these gaps by proposing an integrated framework using deep learning, machine learning, IoT, and explainable AI for early detection, continuous monitoring, severity grading, and treatment recommendation, bridging the gap between research and real-world healthcare applications.

1.4 Research Problem

In an era of growing data-driven, patient-centred solutions in healthcare, the diagnosis and management of chronic conditions such as KOA demands more than standalone diagnostic approaches in the clinical setting. Timely diagnosis, ongoing monitoring, disease severity assessment, and individualized disease management are increasingly recognized as essential elements in enhancing patient outcomes and quality of life. Yet, existing approaches for KOA diagnosis and management are isolated, rigid and lack generalizability. Despite the rapid development of medical imaging, machine learning, and wearable technologies, current approaches remain isolated and do not adaptively integrate multiple streams of data and dynamically change their behaviors in response to the dynamic nature of the disease. This highlights a critical gap in research that needs to not only tackle the limitations of these approaches, but also reimagine them as an integrated, dynamic, and intelligent system for holistic KOA management.

A key limitation is imaging diagnosis, the mainstay of clinical diagnosis for KOA. This is because approaches like X-ray and MRI offer critical insights into the

anatomical changes in the knee joint, such as joint space loss, osteophyte formation, and cartilage degeneration. But these approaches are limited by their intermittent nature, providing only snapshot views of a dynamic process. X-ray images, though affordable and readily available, are largely reflective of bone changes and are ineffective in capturing early cartilage deterioration. While MRI is more complete in capturing soft tissue details, it is costly, less accessible, and not suitable for regular and continuous assessment. Moreover, imaging systems are highly reliant on human interpretation and are generally used only after the disease has reached a critical point, making them less effective for early diagnosis and prevention. The lack of monitoring and predictive capacity in imaging-based systems points to a key challenge: how to bring diagnostic intelligence beyond the clinic and provide continuous monitoring of disease progression.

Similarly, clinical, demographic, and biomarker data have been explored as alternative methods for predicting early KOA. The application of machine learning algorithms to structured data, such as age, BMI, medical history, symptoms, and biochemistry, has shown promise in modelling risk and predicting disease prior to the development of structural changes in imaging. But such approaches have limitations. First, they are usually trained outside of imaging systems, which can lead to less reliable diagnosis due to lack of additional information. Second, clinical models are sensitive to data quality and integrity, with challenges including missing data, class imbalance, and population bias impacting model accuracy and generalizability. Third, many machine learning models are often "black boxes" that are not transparent or interpretable, hindering their adoption in the clinical setting. The research challenge in this area is thus to develop models that not only exhibit high levels of predictive performance but also interface well with other data sources and deliver meaningful and interpretable results.

Although imaging and clinical-based approaches offer important insights, both are inherently limited in nature, as they are static and episodic, and do not consider the dynamic and functional aspects of the knee joint. KOA is a degenerative disease that is affected by daily activities, biomechanical loading, and patterns of motion, but current systems do not provide continuous monitoring of these factors in real-world settings. The emergence of IoT and wearable computing has opened up new

possibilities for continuous health monitoring, especially through methods like VAG signals that focus on vibration signals of the joint during physical activity. But current wearable systems are limited to capturing and analyzing signals offline, rather than integrating with other diagnostic tools. Moreover, systems often rely on cloud processing, which can lead to latency issues, increased energy use, and privacy concerns. The main technical challenge in this area is to create lightweight, efficient and real-time monitoring systems that can be performed on-device, extract relevant biomechanical features and be integrated with other clinical and imaging information to offer continuous monitoring of disease progression.

Beyond disease detection and monitoring, there is also a significant need in the area of severity grading and treatment support. Existing systems for severity assessment, mainly the KL scale, are heavily reliant on radiography and often neglect other common clinical signs and individual factors. This leads to a lack of understanding of individual disease progression and variability. Additionally, current systems often do not go beyond predicting severity to offering clinical advice. The subsequent treatment of decision-making is often done by clinicians independently of the prediction systems. This results in a "gap" between diagnosis and treatment, which limits the effectiveness of decision support systems. The research problem here is to build an integrated system that uses several data sources to perform severity grading and also provide appropriate, personal, and contextual treatment recommendations that are in line with best practice.

These limitations culminate in the more general research problem: the lack of an integrated, multi-modal, and adaptive system that can holistically predict, monitor and manage KOA. Radiological systems lack ongoing monitoring and early warning capabilities; clinical models are stand-alone and often not interpretable; wearable technologies are disparate and not integrated into diagnostic processes; and treatment decision support is not linked to predictive models. Across all areas, current methods lack a comprehensive understanding of the disease and are thus limited in their impact on clinical practice. Current approaches fail to offer integrated, dynamic, and patient-centered solutions that can address the entire KOA management cycle, including early diagnosis, monitoring, and intervention.

This study seeks to overcome these limitations by presenting a multi-modal approach that combines deep learning for image analysis, machine learning for clinical

prediction, IoT for real-time monitoring and fusion for severity assessment and treatment recommendation. The integration of these elements into a cohesive framework aims to connect technological innovations with clinical applications. The new approach focuses on not just performance and accuracy but also interpretability, scalability, and practicality. In conclusion, the study aims to shift KOA management from a reactive, isolated approach to a proactive, continuous, and smart system supporting clinicians and engaging patients, setting the stage for the next wave of adaptive healthcare systems.

1.5 Research Objectives

1.5.1. Main Objective

The main goal of this study is to design an advanced multi-modal AI system that predicts, grades, monitors and supports KOA treatment by integrating medical images (X-ray, MRI), clinical and demographic information, and real-time biomechanical readings from IoT wearable devices.

This system seeks to improve early detection, precise severity classification, monitoring, and decision support with the use of deep learning, machine learning, and XAI. Moreover, the study aims to develop a scalable, interpretable, and real-time decision support system to enhance patient care and assist doctors and therapists in managing KOA

1.5.2. Sub-Objectives

To accomplish the overall goal of developing a data-driven KOA prediction system using clinical, demographic, and biomarker data, the research will address the following sub-objectives:

1. Predicting KOA Using Clinical, Biomarker, and Demographic Data with XAI

- Design and build a machine learning prediction model with structured data such as demographic features, clinical symptoms, and lab biomarkers.
- Develop and fine-tune multiple machine learning algorithms (e.g., XGBoost, Random Forest, Logistic Regression, GradientBoost) and choose the best model according to performance metrics.
- Use Explainable AI methods like SHAP to interpret the importance of features in the model and interpret predictions for individual patients.

- Assess model performance through accuracy, precision, recall, F1-score, and ROC-AUC to evaluate its effectiveness and generalizability.

2. Predicting KOA Using X-ray and MRI Images with XAI

- Build deep learning models with CNNs to classify X-ray and MRI images as having KOA or not.
- Use image preprocessing methods (resizing, normalization, and augmentation) to enhance model performance and generalization.
- Experiment with various CNN models (e.g., EfficientNet, ResNet, DenseNet) to determine the best model for classifying KOA.
- Use Explainable AI approaches like Grad-CAM to highlight critical regions in medical images for the diagnosis
- Assess the model's accuracy with classification metrics and confusion matrices for reliable diagnosis

3. Predicting KOA Severity Using Multi-Modal Data and Developing a Treatment Recommendation System

- Use a multi-modal fusion strategy to integrate imaging (X-ray), clinical and biomarker features for severity prediction
- Develop classification models to predict KOA severity (e.g., KL grades or levels of severity).
- Establish a rule-based and data-driven treatment decision tool based on predicted severity and other patient information.
- Provide tailored output with treatment level recommendation, time of follow-up, lifestyle advice and medical notes.
- Assess the performance of the severity prediction and recommendation system using statistical and logical measures

4. Developing an IoT-Based Knee Health Monitoring System Using VAG Signals

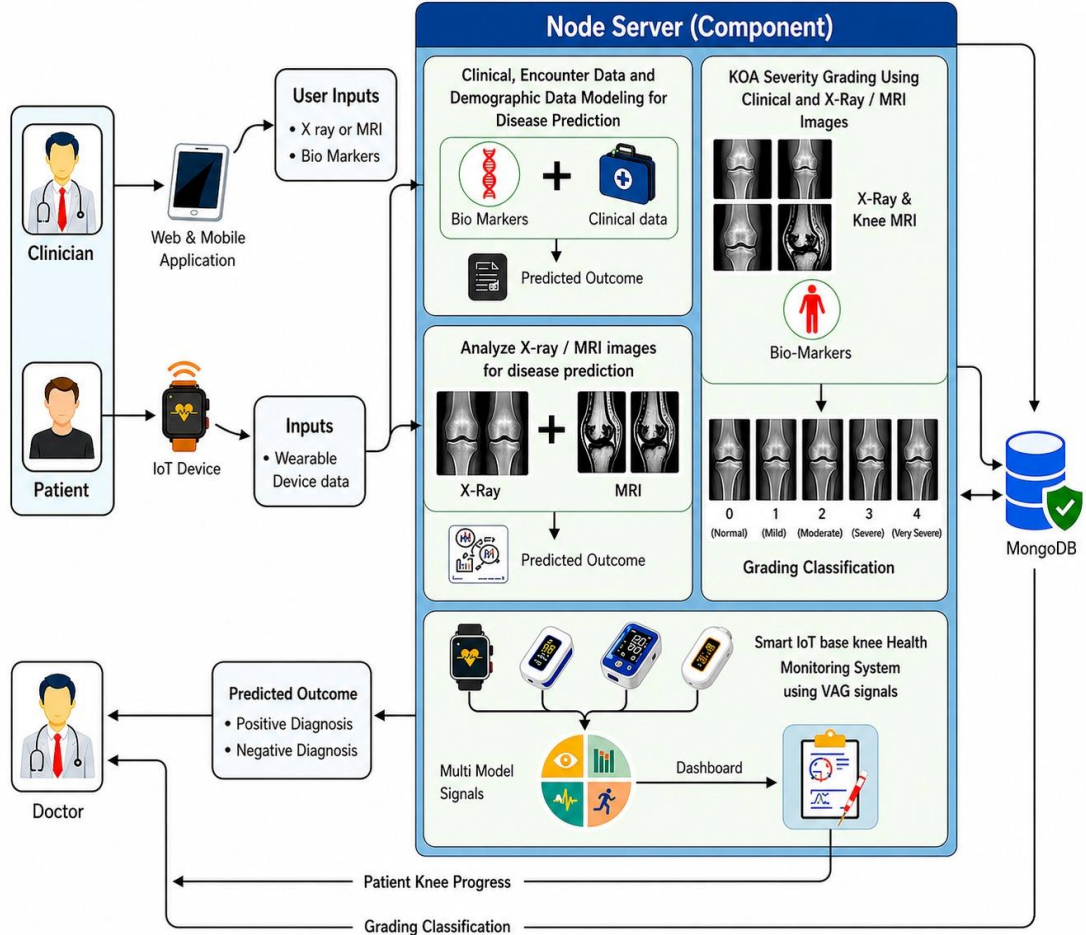
- Create a wearable IoT system to measure VAG signals and other biomechanical characteristics of the knee joint.
- Apply signal processing methods such as filtering, denoising, and feature extraction to extract relevant signal information from sensor data.
- Build efficient machine learning or TinyML models for real-time analysis of knee joint health

- Facilitate long-term monitoring of knee conditions to enable early KOA detection

2. METHODOLOGY

2.1 Methodology

2.1.1. System Overview



The proposed multi-modal KOA prediction and monitoring system is a holistic, end-to-end smart health system that integrates imaging processing, clinical data modeling, and IoT-driven real-time monitoring into a single system. The system architecture involves a Node.js-based server for data processing, machine learning inference, and communication with front-end apps, back-end services, and database storage.

In the first phase, the system accepts data from two main sources: clinicians and patients. Using a web or mobile app, clinicians can submit X-ray or MRI images and patient-specific clinical and biomarker data, such as patient attributes, symptoms, and lab results. Meanwhile, the patient engages with an IoT device that monitors physiological signals related to the knee, including VAG signals, temperature and

motion. Data are then fed into the server in real time, and together with the clinical and biomarker data, create a rich data set that captures the structural and functional state of the knee.

Upon receiving the data, they are fed into multiple processing branches. The clinical and biomarker data are initially fed into a structured machine learning model, which uses demographic and physiological features to determine the probability of KOA. This component uses ensemble methods to model the complex interactions between clinical features and disease manifestations, generating a preliminary prediction.

Meanwhile, the image processing module takes X-ray and MRI images and applies deep learning algorithms to predict KOA-specific features. The images undergo preprocessing steps including normalization, enhancement, and extraction, followed by CNN like YOLO and EfficientNet variants. These networks learn spatial patterns in the images and predict the presence and degree of the disease. X-ray images mainly provide information about joint structure (such as joint space and bone changes), whereas MRI images provide supplementary information about soft-tissue (such as cartilage).

These predictions are then combined in the KOA severity grading component of the system. This module integrates clinical predictions and image outputs using a multi-modal fusion technique, resulting in a severity assessment according to the KL classification (from KL0: normal, to KL4: severe). This integration of structural and individual data provides a more informative assessment of severity than using either type of data alone.

Beyond predictive analytics, the system also features an IoT-based knee health monitoring system, which processes data from wearable sensors. The system extracts vibration features such as RMS, entropy and frequency from VAG signals and uses lightweight models on embedded devices for machine learning. This facilitates real-time assessment of knee joint status, enabling the system to monitor the progression of diseases and identify any abnormalities during routine activities without the need for frequent visits to the clinic.

The pre-processed data and predictions are securely stored in a MongoDB database for longitudinal monitoring and historical analysis. The system further produces visuals via an interactive dashboard, enabling clinicians to track patient conditions, severity evolution, and sensor insights in real-time. The dashboard displays and integrates multi-modal signals, facilitating clinical decision-making.

Finally, the system offers a predicted outcome, which tells the clinician whether KOA is present, or absent, the severity of the KOA and insights specific to each patient. The system presents this information to clinicians via the frontend system, supporting early diagnosis and treatment. The use of imaging, clinical, and IoT data guarantees the system works as a closed-loop, smart system that can support early diagnosis, monitoring, and data-informed decision making in healthcare.

In summary, the system architecture presented here showcases a flexible and scalable approach to healthcare that combines conventional diagnosis with modern AI systems. The integration of multi-modal data, explainable machine learning and real-time clinical monitoring offers a holistic platform to enhance the accuracy, speed and accessibility of KOA diagnosis and

2.1.2. Data Collection and Preprocessing

The design and development of the proposed multi-modal approach to predict, monitor, and grade the severity of KOA required a robust data acquisition and preprocessing strategy to capture variability, standardization, and the real-world nature of the data across all channels. Given that the system fuses imaging data, clinical and biomarker data, and IoT-based sensor data, the data collection strategy was planned to cover multiple aspects of disease to be integrated with a consistent format for seamless integration into the multi-modal fusion system.

The imaging portion of the system uses X-ray and MRI data sets sourced from publicly available databases and clinics, complemented with verified medical images if available. The data includes cases with binary labels for normal and osteoarthritic cases, as well as multi-class labels to reflect the KL grade. Due to differences in image clarity, resolution, and imaging parameters, extensive preprocessing was undertaken. The images were resized to a fixed input size of 224×224 pixels to be compatible with deep learning models. Preprocessing steps like denoising and contrast manipulation (histogram equalization and CLAHE - Contrast Limited Adaptive Histogram

Equalization) were performed to enhance joint definitions. We cropped the image to the ROI to remove unwanted background and to focus on relevant features for the analysis. Data normalization was carried out using standard scaling techniques, and the datasets were divided into training, validation, and test sets with stratified sampling to maintain class balance. Additionally, data augmentation was performed by rotating, flipping, zooming, and adjusting brightness, allowing the model to learn more generalizable features across different scenarios.

Meanwhile, the clinical and biomarker part of the model uses structured medical data from hospital records and enhanced data sets, containing features such as age, gender, BMI, medical history, symptoms, and laboratory tests (e.g., CRP, ESR, cholesterol). Real-world clinical data may contain errors, so a comprehensive data preparation process was applied. Missing data were imputed using suitable methods such as mean, median, or mode of imputation, depending on the variable type. Categorical attributes were encoded using techniques like one-hot encoding and label encoding, and numerical attributes were normalized using z-score normalization to standardize the scale of variables. We also used feature engineering to create new features, such as BMI based on height and weight, and composite scores for symptom severity. Class imbalance was tackled using methods like SMOTE) to balance the number of KOA and non-KOA instances. The dataset was subsequently split into training, validation, and test sets to facilitate thorough evaluation and generalization of the developed models.

The IoT monitoring system introduces a real-time data acquisition system using wearable sensors to measure biomechanical and VAG signals from the knee joint. The sensor data include vibration, joint kinematics, and temperature signals during different activities. Given the lack of large publicly accessible datasets of these signals, data were gathered via controlled experimentation and simulation environments, covering a range of movement patterns and joint states. Preprocessing is essential to feature extraction from sensor signals. Filtering was used to eliminate noise due to external factors and locomotion, and normalization ensured that the signals were within a constant range. Complex signal decomposition techniques such as EMD and EEMD were applied to decompose vibration signals into intrinsic mode functions, allowing the extraction of important features including root mean square amplitude,

peak frequency and spectral entropy. These features reflect subtle joint kinematic features related to cartilage degeneration. To be compatible with embedded systems, the data were preprocessed for efficient inference with TinyML models running on ESP32 microcontrollers. Furthermore, data were also split into a uniform number of windows to enable real-time processing and avoid data drift between training and inference.

To support multimodal integration, all data were formatted to a consistent format and aligned as needed. Clinical predictions, imaging outputs, and sensor features were integrated in a data pipeline to facilitate the fusion model. Rigorous data validation steps were implemented to identify missing, outlier, format and invalid data points. Any invalid or implausible samples were excluded from the analysis to ensure data quality. Additionally, a portion of the data was set aside for reproducibility analysis to ensure consistent assessment of the system's performance throughout the development process.

In summary, the data collection and processing system was structured to ensure that the information from each data stream is accurate, informative, and consistent throughout the system. Through the integration of reliable imaging data, structured clinical data, and dynamic sensor data, the system provides a solid framework for accurate prediction, real-time monitoring, and decision making. This holistic approach not only improves model accuracy but also promotes compatibility and flexibility in clinical settings.

2.1.3. Functional, Non-Functional Requirements

I. Functional Requirements

The system requirements of the multi-modal KOA prediction, monitoring, and decision-support framework outline the system's capacity to function as an integrated, real-time, and adaptive system. The system aims to integrate imaging, clinical prediction, IoT monitoring, and fusion-based severity grading to enable a closed-loop workflow for early detection, ongoing monitoring, and tailored therapeutic advice. Upon system start-up, the framework must ingest multi-modal data comprising X-ray or MRI images, clinical and biomarker data, and sensor data collected via wearable devices. The inputs must undergo modality-specific preprocessing and validation to

ensure uniformity, accuracy, and compatibility in the integrated pipeline; invalid or corrupt inputs will be excluded via a specified error-handling routine.

In the case of the imaging component, the system must support the capture and processing of X-ray and MRI images, ensuring that the images are transformed into the appropriate format for deep learning inference. It must resize the images to a standard size (e.g., 224×224 pixels), normalize them, and apply preprocessing operations such as denoising, contrast adjustment, and region-of-interest cropping to focus on the knee joint. The system must then use the trained deep learning models to infer the presence or absence of KOA and, if present, the severity of the condition based on features extracted by the models. Inference must yield prediction labels, confidence levels, and, in the case of explainability, visual explanations (e.g. heatmaps) that indicate the regions of interest used to predict the outcome. These results should be produced within reasonable time constraints to enable real-time or near-real-time applications.

In parallel, the clinical and biomarker prediction module must receive structured data inputs from the patient, such as age, symptoms, and lab results. The application needs to check the integrity and format of the input data, pre-process the data (including encoding and normalization), and use a trained machine learning model to predict the presence or risk of KOA. The model's output should include the class label and probability scores to allow clinicians to assess the risk. It must also incorporate explainable AI features, including feature importance, to explain how the variables used in the model impact the prediction. This enhances transparency and helps with decision making.

Simultaneously, the IoT monitoring system must monitor and analyze sensor data related to knee joint movements, such as vibration, motion, and temperature. Wearable sensors must capture raw signals in real time and send them to an embedded processor for preprocessing, including noise removal, data normalization, and segmentation. The system must process the signals to extract relevant features using sophisticated signal processing approaches and leverage lightweight machine learning algorithms to identify anomalies or patterns suggesting KOA development. The inference must be conducted on device, if possible, to support real-time and offline usage. The system

must also collect time-series data for trend detection, enabling continuous monitoring of the patient.

A key functional requirement is the integration of the results of all the components into a fusion module. The fusion model must integrate predictions from imaging, clinical, and sensor modalities to provide a holistic evaluation of the patient's status. This involves assessing KOA severity using the Kellgren-Lawrence scale and using information from multiple data sources to enhance accuracy and robustness. The fusion approach must be dynamic, enabling the system to generate outputs despite missing input, by incorporating fallback mechanisms.

Once the severity has been determined, the system needs to trigger the treatment recommendation module, which provides non-prescriptive, patient-specific recommendations based on the assessed severity level and other patient characteristics. This module must consider factors like severity grade, BMI, symptoms, and clinical signs to recommend treatment level, follow-up schedule, lifestyle changes, and clinical factors. The system must provide actionable and easily digestible recommendations, assisting clinicians and patients in understanding recommended actions without overriding clinical decision-making.

The system must facilitate communication between the backend and frontend components, allowing users to enter information, upload images, and view predictions through a single system. Backend services need to handle data processing, model inference, and inter-component communication, while maintaining security of patient data. It must record predictions, inputs, and monitor data for audit, reproducibility, and retrospective analysis.

Overall, these functional requirements define a sophisticated, multi-modal system for health care offering detection, prediction, monitoring, and decision support within an adaptive framework. The system's components work toward their functional goals while working together to improve diagnosis accuracy, enable monitoring, and offer informative, patient-focused insights.

II. Non – Functional Requirements

The non-functional requirements for the multi-modal KOA prediction, monitoring and decision-support system include performance, accuracy, reliability, scalability, usability, security and explainability across the various system components, such as the imaging-based diagnosis, clinical prediction, IoT-based monitoring, and fusion-based severity grading with treatment recommendation. These non-functional requirements ensure the system performs well in practical healthcare settings, ensuring trust, safety and scalability.

The imaging-based prediction component requires high accuracy and performance. The deep learning models should provide high accuracy in classifying KOA and its severity, with comparable performance between X-ray and MRI images. Preprocessing and inference workflows should maintain structural integrity and extract features accurately, with minimal impact from noise and variations in image acquisition. The system needs to be fast to enable near real-time inference, especially for clinical decision making. The system should be robust against variations in quality, resolution, and imaging conditions. The system needs reliability controls to reject invalid images, preventing the system from making decisions on invalid data. Finally, interpretability must be provided through means like heatmaps to allow clinicians to comprehend and verify model predictions.

For the clinical and biomarker prediction system, non-functional requirements include data integrity, scalability, and interpretability. Efficient processing of structured data is essential, with consistent data preparation, normalization, and encoding across different data sources. Models must have high accuracy and generalizability to different patient cohorts with minimal bias. The system should be able to handle missing or partial data; therefore, appropriate data imputation must be applied without significant loss in performance. Explainability is crucial in this component; therefore, feature importance analysis should be provided to ensure transparency of predictions and facilitate acceptance by clinicians. The system must be scalable and support future integration of new clinical factors or biomarkers without significant re-engineering to remain adaptable to new research and datasets.

The IoT-based monitoring component presents stringent real-time, efficiency, and reliability requirements. Wearable devices are deployed in a compute-constrained

environment; the system must enable lightweight and low-power signal processing with embedded machine learning models. It should perform inference quickly to allow real-time monitoring and early detection of knee joint abnormalities. Features extracted from the data should be reliable against noise and other artifacts, as well as changes in the environment, such as temperature. The system must also be resilient to temporary sensor malfunctions or data loss, with redundancy to ensure uninterrupted monitoring. Historical data needs to be saved securely and efficiently, allowing for temporal analysis without high storage or data transfer requirements. It must also be able to work offline, allowing monitoring to continue despite loss of connectivity.

The fusion, severity, grading and treatment recommendation must be accurate, reliable, and interpretable. The fusion system must integrate the results from the imaging, clinical, and sensor-based components to generate accurate severity predictions that correspond to the Kellgren-Lawrence grades. It must handle partial inputs (e.g. when one or more of the modalities are not accessible) and employ fallback mechanisms to ensure continuous operation. The system must provide actionable treatment recommendations in a meaningful, understandable, and unambiguous way, aiding decision-making. The fusion process must also be explainable, so users can understand how the system combines various inputs to produce a prediction and recommendation. This promotes transparency and instills confidence in clinicians and patients.

Overall, the system must be usable and accessible. It must have a user-friendly interface to allow clinicians and patients to enter data, upload images, and interpret the results without technical training. Results need to be communicated effectively, with suitable visualizations and explanations. It should fit with current clinical workflows, without interfering with patient care. The system should be scalable to accommodate more data, users, and extensions, such as other data types or more complex predictive models.

Non-functional requirements include security and privacy, due to the sensitive nature of health data. Data must be stored and transferred securely using encryption, safeguarding against data breaches and accidental loss. It must adhere to applicable data protection laws and ethical guidelines and have robust access controls and usage policies for data. Consent must be sought for the collection of data, especially ongoing

monitoring using wearable devices, and safeguards need to be implemented to ensure that personal data is not divulged without permission.

Finally, it must be reliable and maintainable. The system must be highly available, with high uptime. The system must be modular to enable easy upgrades, model retraining and maintenance without full system redesign. Monitoring and logging should include performance tracking, anomaly detection, and debugging and improvement processes.

In summary, these non-functional requirements ensure that the proposed system provides accurate, robust, secure and interpretable KOA prediction and monitoring. Through performance, scalability, usability, and ethical considerations, the system will operate as an effective healthcare solution in the real world, assisting both doctors and patients with knee osteoarthritis.

2.2 Commercialization Aspect of the Product

There is an opportunity in the burgeoning digital health market for the commercialization of smart healthcare systems for the prediction and management of KOA. With a growing need for early diagnosis and continuous monitoring, along with customized treatment plans, there is a clear market opportunity for integrated systems that extend beyond hospital evaluations. Our research presents a multi-modal approach that integrates imaging diagnosis, clinical and biomarker-based prediction, IoT-assisted wearable monitoring, and fusion-based severity assessment and treatment recommendations to provide a holistic and patient-centric approach.

The system has significant market potential given the large number of people worldwide affected by KOA, especially older adults and those with lifestyle-related risk factors. Early diagnosis and monitoring may lead to cost-effective long term treatment and better outcomes, making it appealing to hospitals, diagnostic laboratories, and telemedicine services. A mixed business model can be employed, offering subscription-based software services to healthcare institutions and integrating with wearable devices for patients.

Collaborations with healthcare providers, wearable device producers, and telehealth platforms can help deploy and expand the solution. The system's modular architecture enables future integration of more data sources or other musculoskeletal conditions,

providing flexibility. In summary, this system can revolutionize KOA care by moving from reactive to proactive, data-driven, and accessible health care.

2.2.1. Market Potential

KOA is a common degenerative joint disorder, affecting millions of people globally, especially the elderly. The growing prevalence of KOA, together with the escalating healthcare demands and rising costs, and the lack of access to sophisticated diagnostic services, has led to a growing need for cost-effective and novel healthcare technologies. Conventional diagnostic techniques including X-rays and MRI are costly, require expert interpretation, and are generally limited to hospital settings making them unsuited for on-going monitoring of disease progress. This multi-modal system overcomes these challenges by providing a cost-effective, convenient and continuous solution for early diagnosis and sustained monitoring. The wearable IoT system combined with artificial intelligence offers real-time monitoring of a patient's health, making it extremely valuable to the health sector, rehabilitation and aged-care industries, and individual patients looking for home-based health monitoring options.

2.2.2. Business Model and the Revenue Stream

The business plan to monetize the proposed system can be best achieved with a combination of B2B and B2C approaches. The B2B model can involve the system being sold to hospitals, clinics, and physiotherapy clinics as a comprehensive system for monitoring and diagnosis. Healthcare facilities can access the system via subscription services that offer powerful analytics, dashboards, and patient management features. Meanwhile, the B2C model allows direct interaction with customers through the provision of wearable devices, and possible subscription services for cloud storage, real-time notifications, and health analytics. Further income can be generated through licensing the AI models, offering API services to other companies providing health care applications and services, and making data-driven insights available for research while adhering to ethical data use practices. This multi-faceted business model guarantees profitability and increased market reach.

2.2.3. Scalability & Adaption

The system has a scalable design that allows it to be used in a variety of contexts and by multiple users. The use of cloud computing eases the processing of large data sets, and supports concurrent use by several users or sites. The system's modular

architecture allows for independent upgrades and extensions of its various modules (such as image models, clinical prediction models, and IoT monitoring modules) without impacting system performance. Additionally, the use of wearable IoT devices allows the system to be used anywhere and enables remote monitoring and home-based care. Its simplicity, along with real-time analytics and visualization, facilitated by web and smartphone applications, ensures user acceptance among clinicians and patients. These capabilities support mass deployment of the system and enable it to be used in settings with both high and low resources.

2.3 Testing and Implementation

2.3.1. Implementation

2.3.1.1. Imaging-Based KOA Prediction Module:

The image analysis component of the system aims to automatically detect and grade KOA from X-ray and MRI images. The implementation starts with a preprocessing step where the image is resized to 224×224 and normalized to suit the input size of deep learning models. Preprocessing also involves denoising and contrast adjustment to enhance clarity of the joint surfaces, and region of interest extraction to focus only on the knee joint region. We use a transfer learning approach with pre-trained deep learning models, such as EfficientNetB0 and YOLOv8 classification models. These models are adapted to a labeled dataset with binary (Normal vs Osteoarthritis) and multi-class labels (KL grading). Cross-entropy loss, Adam optimizer, and early stopping are employed to avoid overfitting, and data augmentation (rotation, flipping, brightness adjustment) enhances model robustness. To improve interpretability, the pipeline also incorporates EAI methods like Grad-CAM, LIME and SHAP, which highlight the regions of the image most relevant to the prediction. Inference involves feeding an image to the model and receiving a predicted label, confidence levels, and (optionally) heatmap visualizations. The trained model is integrated into a Python inference pipeline, which is connected to the backend system to enable real-time prediction via API.

2.3.1.2. Clinical and Biomarker Prediction Module:

The clinical prediction module is tasked with early KOA diagnosis using patient data, such as demographic, symptom, and biomarker data. Data are first verified and processed, including imputation of missing data, one-hot encoding of categorical data,

and standard scaling of numerical data. The model implementation tests several machine learning techniques such as Logistic Regression, Random Forest, Gradient Boosting, and XGBoost. Models with the best performance were Gradient Boosting and XGBoost and were chosen as final models. The training pipeline incorporates hyperparameter tuning, cross-validation, and model evaluation using metrics like accuracy, precision, recall and F1-score. To enhance model performance, class imbalance is handled using methods such as SMOTE, with equal representation of KOA and non-KOA patients. To enhance interpretability, SHAP is used to measure feature importance, thus enabling clinicians to get insights into how parameters like BMI, age and biomarkers contribute to the predictions. The model is serialized and integrated into a Python-based back-end service, which produces predictions in real time based on user-input data.

2.3.1.3. IoT-Based Knee Monitoring Module:

The IoT monitoring module provides real-time knee joint monitoring via wearable sensors. VAG signals and motion data are monitored via sensors (such as accelerometers and microphones) placed around the knee joint. Real-time processing of raw signals is performed using embedded systems for efficient and low-power processing. The pre-processing stage consists of filtering, normalization, and windowing of the signals into a constant length. Sophisticated signal processing methods, such as EMD and EEMD, are used to break down the signals into intrinsic mode functions. Biomechanical features of the knee joint, such as RMS amplitude, peak frequency, and spectral entropy, are then extracted from these components. These features are then used as input to lightweight machine learning models designed for TinyML on ESP32 microcontrollers. These models classify or detect anomalies in real-time to detect patterns related to KOA. Continuous recording of time-series data is performed for longitudinal monitoring, and summaries are sent to the back end for further processing with other modules. The design is offline enabled, enabling monitoring in the absence of an internet connection.

2.3.1.4. Multi-Modal Fusion and Severity Grading Module

KOA severity prediction module was created to estimate severity levels by integrating imaging data with clinical information. For the model built on imaging data, EfficientNetB0 was employed to learn from preprocessed X-ray images with size

224×224 to capture critical radiographic information like joint space narrowing and osteophyte presence. The output is severity estimation consistent with the KL grade. Also, the clinical model based on XGBoost was established, taking advantage of the clinical data of patients, such as age, BMI, symptoms, and biomarker information. The preprocessing steps involved dealing with missing values, categorical variable encoding, and normalization.

Late fusion was conducted to integrate predictions from the two models. Prediction probabilities from EfficientNetB0 and XGBoost were aggregated to form the final prediction on severity levels according to the Kellgren-Lawrence grading system (KL0-KL4).

2.3.1.5. Treatment Recommendation Module:

The treatment recommendation module offers non-prescriptive, personalized recommendations based on severity prediction and individual characteristics. Our solution employs a rule-based approach that translates severity grades, clinical factors, and patient attributes into treatment recommendations. The inputs (KL grade, BMI, symptoms, medical history) are assessed to provide outputs such as the level of treatment recommended (lifestyle modification; physiotherapy; medical treatment), recommended follow up, and suggested lifestyle advice. The system provides recommendations in a transparent and understandable way, aiding both doctors and patients in the decision-making process. The module is developed to simulate medical reasoning processes and can be easily replaced with recommendation systems based on machine learning. It is integrated into the backend system, allowing communication with predictions and front-end systems.

2.3.1.6. System Integration & Deployment

The system is built with a modular architecture that consists of frontend, backend, and machine learning modules. The backend is built using Node.js and Express framework, which manages API calls, data manipulation, and module interaction. Machine learning models are developed in Python and executed in a child process to allow communication between Python and Node.js. The frontend is built with cutting-edge web frameworks, enabling users to upload images, provide clinical information, and receive predictions and recommendations through a user-friendly interface. Frontend and backend data exchange is facilitated by RESTful APIs for secure and efficient

communication. It is designed for deployment on various platforms, such as local servers, cloud servers, and embedded systems. Models are dynamically loaded and stored, allowing for updates and retraining with minimal impact on system operation. Error handling and logging are provided for robustness and maintainability.

2.3.2. Testing

2.3.2.1 Functional Testing

Functional tests were performed to ensure that all modules of the multi-modal KOA diagnostic system meet their requirements. For instance, the imaging module was evaluated by feeding X-ray and MRI images to test whether the module performs preprocessing (resizing, normalization, ROI extraction) correctly and produces predictions regarding the existence and severity of KOA with accuracy. Explanations such as heatmaps were evaluated to check whether there is appropriate highlighting of joints' regions.

The clinical prediction module's functionality was checked using data about patients in a structured format to ensure proper handling of missing values, encoding, and normalization. Moreover, XGBoost models were evaluated based on correctness of predictions and probability scores produced, as well as SHAP outputs' interpretability.

Finally, the IoT monitoring module was tested to validate correctness of the following procedures performed on signals: acquisition, preprocessing (filtering, segmentation), and feature extraction. The TinyML model was evaluated for proper detection of abnormal patterns on device level in real time.

The severity prediction model was assessed for proper integration of EfficientNetB0 (image classifier) and XGBoost (clinical model) results using the fusion method. Validation was done to ensure that the results were generating proper values of the KL grade from 0 to 4.

Similarly, the treatment recommendation model was assessed using varying degrees of severity to validate the mapping of outputs including treatment degree, follow up time, and living style recommendations. The end-to-end testing ensures that proper inputs are mapped to proper outputs through all models.

2.3.2.2 Performance Testing

Performance tests were performed to determine the accuracy, speed, scalability, and responsiveness of each module in the multimodal KOA framework. The aim was to verify the reliability of the system in making predictions while ensuring good performance in terms of computational time.

For the image processing modules, the performance analysis was carried out by measuring common classification metrics such as accuracy, precision, recall, and F1-score on the training, validation, and test sets. The EfficientNetB0 model showed consistent results in the multi-class severity classification task, while the YOLO classifier showed excellent accuracy in binary classification (Normal/Osteoarthritis). The learning curve and confusion matrix were also analyzed to validate the convergence of the model and its performance on each class, especially for distinguishing between adjacent KL classes (e.g., KL1/2).

The performance of the clinical prediction model was examined on structured datasets, resulting in high accuracy and good generalization ability for new data. Cross-validation was conducted to examine the stability of the model, while the performance measures indicated that the model made consistent predictions for patients. The feature importance evaluation also verified that variables such as body mass index, age, and biological markers played a significant role in influencing the model's decision-making.

The fusion-based severity prediction module was validated by comparing its performance with other modules. The outcomes revealed that integrating the results of imaging and clinical modules led to increased prediction of accuracy and decreased uncertainty, particularly for patients whose conditions were difficult to classify into severity categories. The fusion method provided higher consistency and reliability than separate predictions made from each modality.

For the IoT-based monitoring module, the performance was evaluated based on real-time data processing and computing efficiency. Signal pre-processing, feature extraction, and TinyML inference were all carried out using embedded systems to guarantee timely data processing and minimize computation burden.

Performance tests were conducted at the system level to assess performance related to the processing of information on the backend and its speed. Performance was evaluated by analyzing the speed of API calls to ensure that predictions and recommendations are delivered within reasonable time intervals. This test proved that the system can efficiently handle a large number of requests.

In conclusion, performance tests have shown that the proposed system is capable of balancing prediction accuracy and execution efficiency, thus making it fit for implementation in clinical settings.

2.3.2.3 Robustness Testing

The robustness test was carried out to check the system's stability under different conditions and nonideal inputs. To test the imaging part of the system, poor-quality images, as well as those containing noise and partial occlusions, were used to guarantee the stability of the system and the safe rejection of wrong inputs.

The clinical model was tested using an incomplete set of data to validate that it can handle missing data and still make predictions accurately. Different distributions of the input data were used to verify the generalizability of the model.

To check whether the IoT system can function under noise and disturbances in sensors' readings, such signals were simulated in the environment to test for feature extraction and predictions. In addition, a disruption in the transmission of signals for a while was simulated. The fusion module was tested with missing modalities (e.g., lack of imaging and sensors' data).

2.3.2.4 Integration Testing

Integration testing was performed in order to test whether all the different elements of the multi-modal KOA system integrate well with each other as a whole. It involved testing how the frontend, backend, ML algorithms (imaging, clinical, fusion), as well as the IoT monitoring system interacted with each other.

The API functionality between the frontend and backend was tested to ensure that there was proper communication of data from the former to the latter in terms of images being uploaded, clinical data, and predictions outputted. The interaction between the Node.js backend and the models written in Python was tested in order to ensure that

they worked properly with each other in terms of the prediction being outputted properly in the form of JSON response.

Testing was also done on the combination of the two different models of EfficientNetB0 and XGBoost in order to check whether they worked in tandem to give a prediction in the fusion model. End-to-end testing of the system was done using full data inputs and checking if the outputs and predictions were stable across all inputs.

3. RESULTS

3.1 Results

Imaging-based KOA prediction component performance was assessed by testing the models based on both the X-ray and MRI datasets, which utilized deep learning algorithms for classifying whether osteoarthritis is present in the image. YOLO-based classification algorithm for X-rays performed exceptionally well with validation accuracy close to 91% and test accuracy above 90%. This indicates the capacity of the algorithm to distinguish between osteoarthritic and normal knee images. Training and validation loss values also confirmed convergence without over-fitting. As for MRI-based model, it performed relatively poorly with validation and test accuracy being close to 83%. This can be explained by the increased complexity of MRI images compared to X-rays. Confusion matrix analysis showed that the algorithm built for X-rays produced very few false negatives, which is vital when performing medical diagnoses and not missing positive results. Besides, the use of Explainable AI algorithms like Grad-CAM, LIME, and SHAP proved that the models are focusing on relevant features including joint space narrowing for X-rays and soft tissues for MRI.

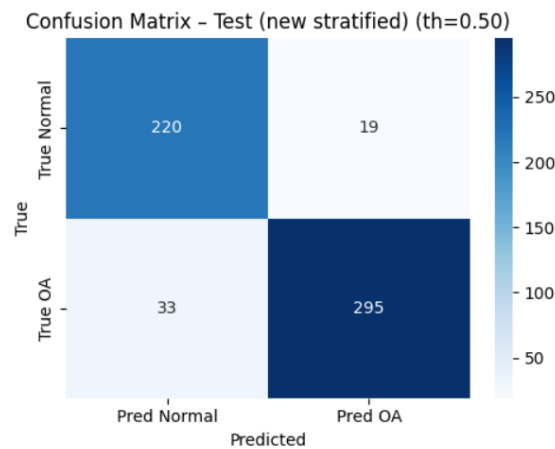


Figure 3 - Confusion Matrix of the Imaging Component (X-Ray)

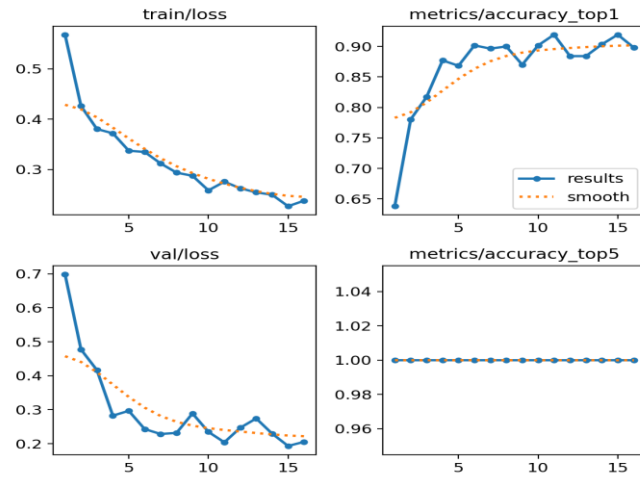


Figure 4 - Results of the X-Ray Prediction Module

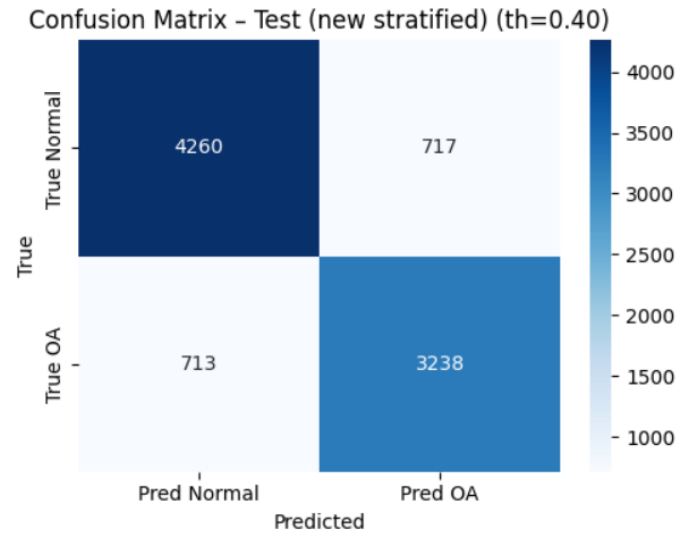


Figure 5 - Confusion Matrix of the Imaging Component (MRI)

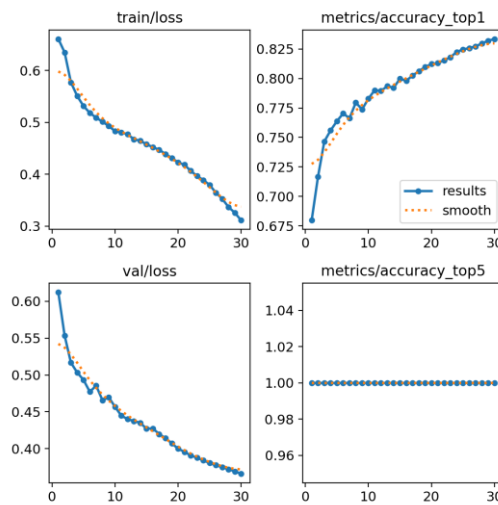


Figure 6 - Results of the MRI Prediction Module

For evaluating the clinical and biomarker-based prediction component, multiple supervised machine learning algorithms were used including Random Forest, Logistic Regression, KNN, Gradient Boosting and XGBoost. Among the models, the Gradient Boosting model performed best in terms of accuracy, reaching a training accuracy of 89.1% and testing accuracy of 89.0% indicating little variation between both training and testing accuracy, thus minimal overfitting. This is an important feature as it would enable the model's generalizability in real life clinical use. In terms of accuracy, Random Forest performed similarly to Gradient Boosting but experienced a slight reduction in testing accuracy (88.5%) compared to training accuracy (90.5%), indicative of minor overfitting. XGBoost, despite its reputation as one of the strongest algorithms, exhibited a slightly reduced testing accuracy (87.5%) compared to Gradient Boosting and Random Forest, emphasizing how important it is to have good dataset characteristics for achieving desired performance from specific models. Moreover, the Learning Curves confirmed the balance between bias and variance of the chosen Gradient Boosting model and showed increasing stability in its validation accuracy with respect to growing dataset size. Confusion matrix further justified the model's effectiveness while normalization, encoding and SMOTE preprocessing helped improve its robustness.

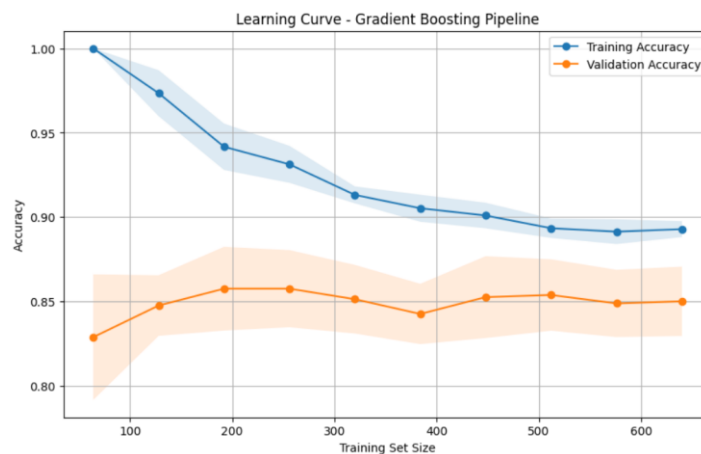


Figure 7 - Learning Curve of the Final Clinical Model

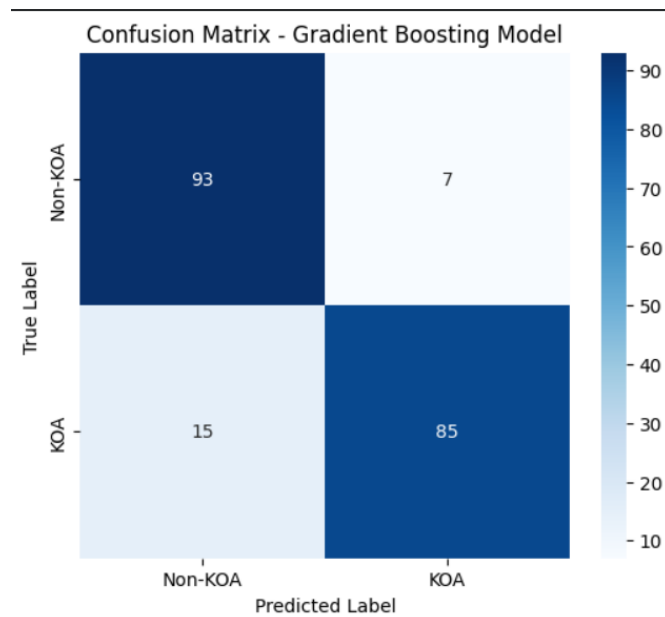


Figure 8 - Confusion Matrix of the Final Clinical Model

The KOA severity prediction part was developed through the use of EfficientNetB0 for the classification of images and XGBoost for clinical data. Both were then combined in order to make the final predictions. In particular, the EfficientNetB0 model managed to reach an accuracy rate of 81.34% in training and 78.00% during validation and testing at a value of 80.25%, suggesting that the model can be used effectively for multi-class severity classifications. From the confusion matrices, it was seen that the model performed well in recognizing severe cases, such as KL3 and KL4, with values of up to 98.73% precision and 96.27% recall. Nonetheless, slight inaccuracies were found when trying to recognize adjacent classes such as KL1 and KL2, which is normal considering how similar the two cases may be in earlier stages. The clinical XGBoost model managed to perform better in terms of accuracy, achieving scores of 88.44%, 86.88%, and 89.00% on training, validation, and test respectively, implying good results in structured data prediction. By applying late fusion, the model showed better results in overall predictions. The output mapped the predictions according to the Kellgren-Lawrence grading scale (KL0-KL4).

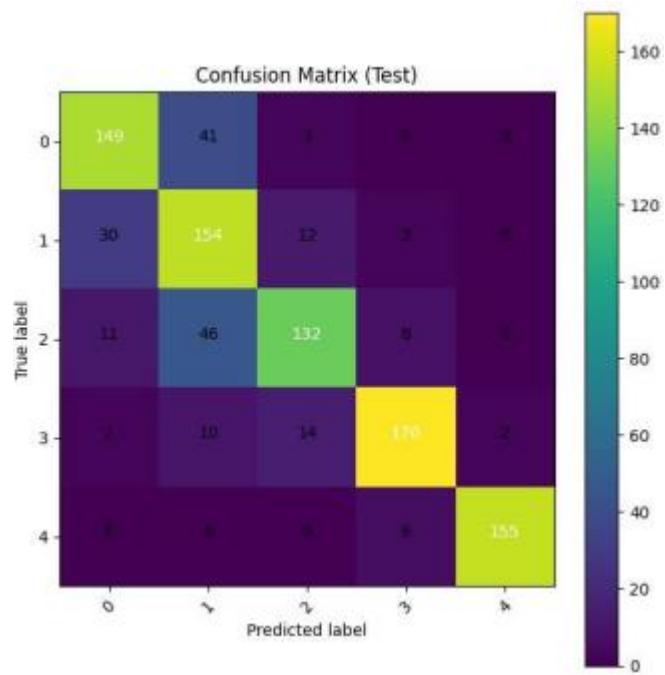


Figure 9 - EfficientNetB0 Model Confusion matrix

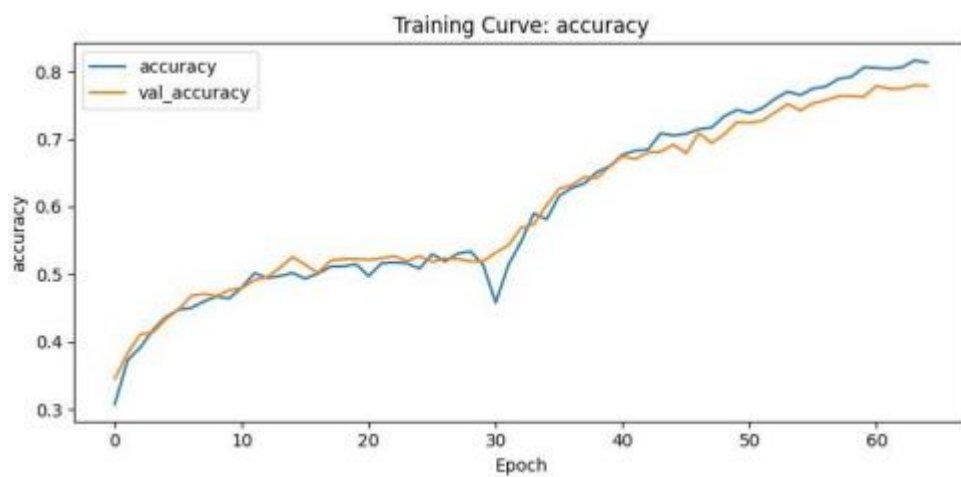


Figure 10 - EfficeintNetB0 Model Accuracy Curve

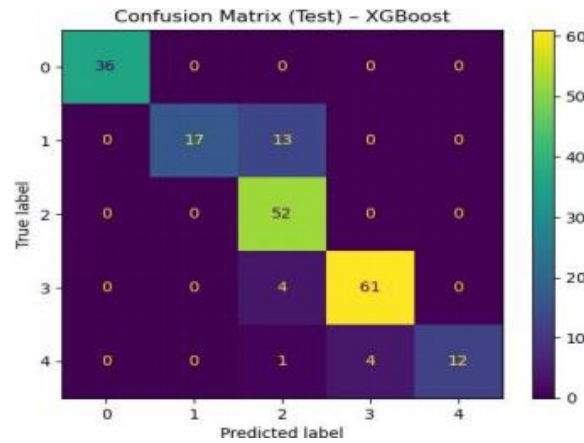


Figure 11 - Confusion matrix of the XGBoost clinical model

Classification results of the IoT-based knee monitoring module based on vibrations and temperature were carried out using a decision tree classifier, and a classification was performed on four different severity levels, namely: Normal, Mild, Moderate, and Severe. The accuracy of classification was very high for two severity levels: Normal and Severe. From the confusion matrix, it was noted that the accuracy of classifying Normal samples was 156, while for Severe samples, it was 139. There were minor inaccuracies noted between the Mild and Moderate, and the Moderate and Severe cases, which can be attributed to the fact that there are commonalities among the features of the different levels of severity. This classifier provided accuracy in both training and testing sets, thus proving that it generalizes well for unknown cases. Finally, the IoT-based knee monitoring module was able to perform accurate real-time signal processing and feature extraction.

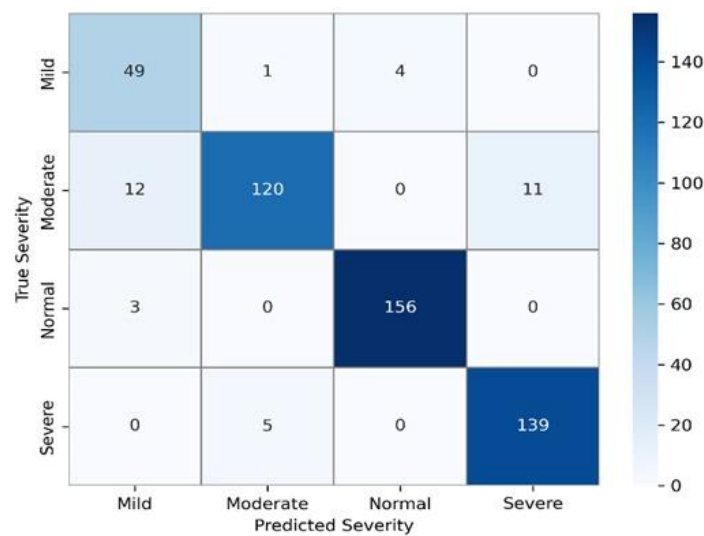


Figure 12 - Confusion Matrix of the Knee Monitoring Module

Lastly, the module that generates treatment recommendations was analyzed according to its capability to produce adequate outputs that correspond to different predicted levels of severity and inputs provided by individual patients. In all cases, the output recommendations were adequate; they included treatment levels, recommended intervals between follow-ups, lifestyle modifications, and additional clinical recommendations. The outputs were clear and understandable, and they coincided with clinical practice guidelines. This makes the integration of the module possible as the step towards creating the whole functionality of the proposed system.

To sum up, the tests conducted show that each part of the system works effectively within its own area. However, the usage of several types of information – from imaging, clinical sources, and IoT technology – helps to improve the performance of the whole system.

3.2 Research Findings

From the results obtained during the experimental assessment of the proposed dual-modality system for KOA prediction, we can conclude that deep learning is quite efficient in classifying this disorder based on medical images. Indeed, YOLO-type models proved to be quite capable of detecting KOA from radiographic and MRI images, showing great efficiency in terms of accurate classification. The performance of the X-ray-based model was better than that of its MRI counterpart, which suggests that radiographs are efficient tools for visualizing the changes associated with KOA (joint space narrowing, bone abnormalities).

On the other hand, while less accurate, the MRI model allows gaining additional complementary information by detecting structures, such as cartilage and ligaments, invisible through X-rays. Thus, the use of the MRI model becomes essential for early detection of changes associated with KOA that do not affect bone tissues yet. In conclusion, the presented research proves that although X-ray provides an effective base for diagnostics, adding MRI to the process enables obtaining comprehensive information related to both types of tissues.

In addition, the models' low rate of false negatives becomes a significant strength that should be considered when implementing them in clinical practice. Minimization of false negatives decreases risks associated with underestimation of the disease, which makes X-ray model especially appealing due to its reliability in detecting KOA. As for

Grad-CAM, LIME, and SHAP, their application made it possible to confirm that the proposed models are focused on clinically relevant areas of the knee joint. It can be explained by the fact that the use of explainable AI allows establishing a connection between machine learning approaches and medical science.

The insights gained from the clinical and biomarker-based prediction module clearly indicate the high possibility of using non-image information to detect KOA at an early stage. The analysis shows that demographic variables, clinical signs, and biomarker levels can efficiently predict KOA with high accuracy even without relying on imaging information. Such an approach will be highly beneficial in clinical settings where imaging data may not be accessible, thus enabling the early detection and screening process using easily available patient data.

Another significant insight from the experiment was the efficiency of ensemble techniques in processing structured medical data. The results show that the Gradient Boosting technique produced the best performing model in terms of testing accuracy with the least level of overfitting. This suggests that ensemble approaches are highly effective in capturing non-linear dependencies among medical variables. In addition, the simpler models, including logistic regression and KNN, underperformed compared to ensemble techniques, suggesting that researchers should choose appropriate algorithms when analyzing heterogeneous medical data. Notably, model complexity did not play a significant role in improving the predictive power of the machine learning algorithm since XGBoost underperformed despite its sophisticated nature.

The incorporation of Explainable Artificial Intelligence methods is another factor that makes this module more practical. SHAP explained how important specific features are in making predictions, including such parameters as BMI, age, pain, and inflammation markers (such as C-reactive protein and erythrocyte sedimentation rate). This information corresponds to what is known from the field of medicine and thus demonstrates the practical applicability of the presented approach. Interpretability is important because it allows doctors to better understand their patients' condition. Based on this evidence, one can conclude that the use of clinical data and biomarkers is an excellent tool to diagnose KOA early on.

The results of the severity prediction part show the benefits of integration of both imaging and clinical information via a fusion-based strategy. EfficientNetB0 proved

to be effective at detecting high severity grades (KL3 and KL4) as there is a visible structural difference between these stages of the disease; therefore, it is rather easy to differentiate them. On the other hand, the algorithm had trouble differentiating between adjacent severity grades (KL1 and KL2).

The clinical model was superior to the imaging model as far as its accuracy is concerned, which proves the presence of valuable information within the structured patient dataset. It can be seen that the BMI, pain score, and biomarker levels are very important. At the same time, the use of clinical models alone could be problematic due to the lack of structural joint information.

The combination of imaging and clinical outputs led to increased consistency and reliability in predicting KOA severity. The incorporation of structural data together with patient data allowed for the reduction of uncertainty, especially in marginal cases. This proves that the use of multimodal techniques is more informative than unimodal techniques in predicting KOA severity. Moreover, the final outputs matched the Kellgren-Lawrence grading scale, which suggests the feasibility of applying the proposed system in clinical practice. In conclusion, the results support the notion that model fusion is the optimal technique for predicting KOA severity.

The results obtained from the module for generating treatment recommendations prove that it is highly effective and efficient in transforming the outputs from prediction into useful recommendations. The rule-based module was able to generate treatment recommendations using severity levels and some factors unique to individual patients (e.g., BMI and symptoms).

The most important finding here is the effectiveness of the recommendation generation, which is usually lacking in other AI diagnostic approaches. Providing recommendations like treatment levels, the interval needed between visits, and lifestyle modifications makes the system very useful and practical. Also, the rule-based approach provides clear and comprehensible recommendations for the user.

The major limitation of rule-based modules is their lack of adaptability to more complex cases. To deal with this problem, future improvement could include using machine learning-based recommendation systems. In conclusion, the treatment

recommendation module shows very good results and proves to be very useful and promising as an essential part of the KOA diagnostic approach.

The results from the IoT-based monitoring technique indicate the efficacy of wearable sensor-based prediction techniques in the accurate prediction of the real-time status of the knee joint. The system efficiently utilized vibration-based features as well as temperature measurements to predict the KOA severity levels. RMS value, frequency, and entropy were among the important features recognized by the system.

The confusion matrix showed that the classifier performed outstandingly well when it came to predicting normal and severe conditions since both were clearly differentiable. Some overlap existed between mild, moderate, and severe conditions; this is because of the gradual shift between them. Nevertheless, classification accuracy was high despite the overlapping nature.

One more interesting finding is the high level of explainability of the decision tree algorithm that was applied in the developed component. Contrary to neural networks and other deep learning models, decision trees have transparent decision paths, which is a very important property that helps use this type of algorithms in medical diagnostics because doctors must be aware of the reasoning process. Furthermore, by using IoT technologies, it became possible to monitor patients even when they are not in hospitals, which is an innovative solution compared to existing diagnostic methods based on hospital visits.

3.3. Discussion

Indeed, findings related to the imaging-based component show that the deep learning models, especially those using YOLO-type architectures, can be successfully applied for the diagnosis of KOA using medical images. Since the high level of accuracy and ROC-AUC value shown by the X-ray model implies reliable extraction of joint space narrowing and osteophyte formation, the use of deep learning methods for detecting KOA in X-rays proves to be quite promising and matches other relevant research studies [5], [6]. Nonetheless, the relatively low level of accuracy achieved by the MRI-based model reveals the problem of high complexity associated with the analysis of high-dimensional medical data. In particular, the presence of rich information about soft tissues in MRI images, despite its clinical value, increases the complexity of feature extraction. This finding corroborates other relevant studies that discuss

difficulties related to the application of deep learning to the analysis of MRI images [7], [15]. One major point to be learned from this part is that there is complementarity between the use of X-ray and MRI modalities. Though X-ray acts as an excellent baseline to detect structures, MRI can help identify further details about cartilage and soft tissue degradation. This finding has been validated by multi-modal imaging research [9], where it is pointed out that using more than one modality in medical imaging increases the accuracy of diagnoses. In this way, this component contributes to bridging the research gap identified at the outset as it demonstrates the advantages of dual-modality imaging. In addition, by integrating explainable AI methodologies, this component has improved the interpretability of deep learning models and overcome the drawback pointed out in [8].

Clinical Prediction shows that non-image data can prove extremely effective when used in early KOA prediction. The excellent results demonstrated by Gradient Boosting prove that there is a lot of useful data in demographics, clinical features, and biomarkers. This corresponds with research works [2], [3], [4] that prove that such elements as BMI, age, and inflammatory biomarkers play a great role in the risk of developing KOA. One should also point out the superiority of ensemble learning techniques over traditional machine learning algorithms. The fact that Gradient Boosting performed significantly better than logistic regression and KNN proves that medical data should be taken into account in terms of non-linearity. It is supported by such works as [11], [12], which emphasize that ensemble algorithms generalize better. It is worth mentioning that even though XGBoost is considered one of the most advanced models, it proved to be less efficient than Gradient Boosting. One may conclude that the effectiveness of a chosen algorithm depends greatly on data properties. This becomes even more significant with the introduction of explainable AI, which offers an interpretation that is understandable regarding the insights generated from the predictions of these models. The clinical features used become more relevant and consistent with medical expertise, enhancing credibility and facilitating decision-making. Nonetheless, clinical models, while effective for predictive purposes, lack validation of their structure based on imaging data. This is a major reason why they should be integrated with other modalities as well.

Fusion-based severity prediction module proves the significance of multi-modal information usage for improvement in diagnosis accuracy. The experiments reveal the advantages of using the combination of EfficientNetB0 and XGBoost in the task of severity classification compared to single models. In the case of the imaging model, it was observed that it had a high rate of recognition of high-severity categories (KL3 and KL4), characterized by the presence of clear structural alterations. This finding is in line with the results obtained by previous studies [6], suggesting the excellent ability of deep learning algorithms in recognizing advanced stages of KOA. Still, the mistake between stages (KL1, KL2) can be considered normal in medical image analysis tasks. However, the clinical model showed excellent predictive ability as well, showing that patient-based information is critical to comprehending the progression of the disease. The results corroborate those found in [3], [11], in which the clinical parameters proved to be important factors predicting the severity of the condition. Fusion is capable of incorporating all these strengths into one process, minimizing uncertainty and ensuring higher predictability. Such a strategy confirms the conclusions of the current multi-modal analysis [9], [10], according to which combining different types of data proves to be quite efficient. Most importantly, this aspect contributes greatly towards closing the existing research gap. Moreover, the capability of the fusion model to deal with missing modalities makes it more applicable, thus enabling its application in healthcare institutions despite incomplete data availability.

Firstly, the IoT-based monitoring component demonstrates the increasing significance of wearable devices in healthcare. It is shown that vibration-based parameters and temperature data can be successfully used in monitoring and detecting alterations occurring at the knee joint due to KOA. These findings are consistent with the results presented in [16], [17] suggesting that wearable devices can be effectively used in joint monitoring. Another significant advantage provided by the component under analysis lies in the fact that it allows for constant monitoring of patient condition beyond the walls of hospitals and clinics. It contrasts with conventional methods, which usually require regular visits to hospitals for diagnosis. At that, the proposed IoT solution facilitates prompt identification of any changes in patients' conditions. In that way, it is consistent with research in [18], [19]. Nevertheless, the noted problem with misclassification between adjacent severity levels suggests that sensor-based data alone cannot ensure proper classification of patients' KOA severity level. The other

significant feature is the use of explainable models like decision trees that ensure transparency in the predictions. This becomes even more relevant in case of medical applications since transparency plays an important role. In essence, this highlights how well-suited IoT systems are for continuous monitoring, although they need to be combined with other methods of diagnosis.

The module for generating treatment recommendations is vital for implementing the predictions into meaningful actions. From the findings, it is clear that the rule-based system successfully correlates severity classes and patient characteristics with treatment advice. This solves a major drawback seen in previous literature where many AI models have been developed to predict certain medical conditions but fail to generate any actionable insights. In this regard, the current model is significantly more useful than others due to this feature. Furthermore, the interpretability of the rule-based model is another advantage that makes it more suitable for application in the real world. Interpretability ensures that there is no black-box situation when it comes to understanding how the treatment recommendations are formed. However, one limitation of the model is the use of a pre-defined rule structure, which can be a disadvantage in terms of flexibility. This can be improved in the future by integrating machine learning-based recommendation systems. In summary, this particular component brings closure to the end-to-end process of the entire AI model.

4 SUMMARY OF INDIVIDUAL CONTRIBUTION

4.1. JAYASINGHE J M N S (IT22582942)

First, Jayasinghe was involved in developing a machine learning algorithm that predicted KOA. The algorithm consisted of clinical, demographic, and biological features. The goal of the research was to build an efficient and interpretable framework that could predict the disease's presence with high precision.

The next step was implementing and comparing different machine learning algorithms and choosing the most accurate one. For example, Gradient Boosting proved to be the most appropriate because it showed high generalization performance and could detect non-linear associations in healthcare data. Moreover, the researcher used XAI, such as SHAP and feature importance scores, to interpret the results.

As for the third stage, the researcher dealt with the real-life problem of missing data and feature variance by applying specific methods to enhance the efficiency of the process. Thus, this contribution to the research made it possible to identify patients at risk and facilitate the data-driven decision-making process.

4.2. FERNANDO W D A (IT22223708)

The imaging-based KOA detection part of this project was led by Fernando, who implemented the use of a deep learning model based on X-ray and MRI images. He implemented an entire image pre-processing pipeline including the normalization, enhancement, and regions of interest selection to improve image quality and enhance the output accuracy.

Different deep learning neural networks were investigated, among them are LeNet-5, AlexNet, GoogLeNet, and YOLO. The results obtained from analyzing the different deep learning algorithms showed that YOLO neural network algorithms performed better than other neural networks and thus were chosen as the solution.

Explainable AI algorithms like Grad-CAM, LIME, and SHAP were incorporated into the system to help in interpreting the prediction made by the deep learning algorithm.

4.3. PERERA B B A R (IT22606792)

Perera helped develop the KOA severity prediction and treatment recommendation system which combines the use of imaging and clinical information into one. This required implementing models based on deep and machine learning, including using

the EfficientNetB0 algorithm for severity prediction using images, and building a structured clinical prediction model through preprocessing, including filling up missing values, encoding, and feature scaling.

An important achievement is the creation of a multi-modal fusion system where results from imaging and clinical models are combined by means of the late fusion technique in order to increase accuracy and make predictions more reliable, which made it possible to reliably predict the severity of KOA according to the Kellgren-Lawrence scale.

In addition, a rule-based Treatment Management Support System was designed that provided treatment level, follow-up time frame, life style recommendation, and clinical information in order to create individualized recommendations for each patient.

Finally, Perera also worked on developing the web app as a whole, which includes the frontend and backend components, API integration, and overall connectivity.

4.4. GAMAGE D.M.G.P.K (IT22188472)

The role that Gamage had been assigned was the development of the IoT based knee health monitor for the purpose of obtaining the relevant data in real-time using IoT technology. This consisted of integrating the appropriate sensors including vibration, motion and temperature sensors into an ESP32 microcontroller.

Data preprocessing and feature extraction was part of this contribution which consisted of preprocessing the raw sensor data obtained in order to derive meaningful features including the RMS, Entropy and Frequency Components using data transformation techniques. A decision tree classifier model had been implemented for the purpose of classifying the severity of KOA due to the high accuracy and ease of interpretation of this algorithm.

Moreover, this component consisted of integration and visualization of the IoT based knee health monitor where the IoT device was integrated into the backend server and a visual interface was designed.

5 CONCLUSION

Consequently, the developed approach successfully integrates multimodal data, including imaging, clinical data, and information gathered by Internet of Things sensors, thus establishing a powerful framework for the prediction, severity grading, and continuous monitoring of KOA patients in real-world settings.

The imaging-based component showed the high efficiency of deep learning approaches for identifying KOA. In particular, the YOLO-based model proved to be highly efficient at detecting KOA through both X-rays and MRIs. The higher efficiency of X-ray-based models was driven by their capacity to detect more pronounced anatomical characteristics, while MRI provided additional information about soft tissues. The use of multimodal imaging proved highly efficient.

In its turn, the clinical and biomarker-based component validated the possibility of using non-imaging data to accurately detect early signs of KOA. In particular, ensemble machine learning methods such as Gradient Boosting appeared to be highly efficient in recognizing complex interactions between various clinical factors. It should be noted that the presented approach is highly practical in terms of cost and accessibility.

The severity prediction module, which involved the integration of EfficientNetB0 with XGBoost using a fusion technique, was found to be superior to each model individually in terms of accuracy and reliability. The integration of both structural imaging data and patient-specific clinical information enabled the classification of KOA severity as per the Kellgren–Lawrence scale in a better manner. Therefore, the multi-modal fusion model directly overcomes some of the issues associated with single-modality models.

The IoT monitoring component provided the design of an IoT-based wearable device that can obtain real-time data from the knee joint and continuously monitor the disease. The application of vibration-based data and TinyML models shows the ability of real-time and cost-effective health-care solutions. Thus, this aspect is quite advanced compared to the conventional method used for the diagnosis of such diseases.

Moreover, the treatment recommendation module helped to translate the prediction results into action. It helped to make treatment recommendations based on the prediction results. This included the selection of the treatment level and follow-ups. In summary, the system shows how effective it can be to incorporate multi-modal data, explainable artificial intelligence, and real-time monitoring when it comes to detecting and managing KOA. This paper helps advance intelligent health care systems by developing an entire framework that allows not only early detection and continual monitoring but also severity assessment and personalized treatment.

The next step could be increasing the size of the database to make it more diverse and enhancing the process of determining the severity in the early stage of the disease. Moreover, one can develop more advanced deep learning models and adaptive recommendation approaches.

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GLOSSARY

Knee Osteoarthritis (KOA): A degenerative joint disease affecting the knee, characterized by cartilage breakdown, pain, stiffness, and reduced mobility.

Machine Learning (ML): A subset of artificial intelligence that enables systems to learn patterns from data and make predictions or decisions without explicit programming.

KL Grade (Kellgren–Lawrence Scale): A grading system for the severity of osteoarthritis ranging from KL0 (healthy) to KL4 (very severe).

X-Ray: A method of medical imaging that visualizes bone structures, often used to identify joint space narrowing and bone alterations in KOA.

MRI (Magnetic Resonance Imaging): A high-resolution method of imaging that creates detailed images of soft tissues like cartilage, ligaments, and meniscus.

Deep Learning: A form of machine learning where neural networks with several layers are trained on big data sets to detect patterns automatically.

YOLO (You Only Look Once): The chosen framework for real-time object detection and classification in this research for KOA identification.

EfficientNetB0: A convolutional neural network model designed for maximum accuracy and computational efficiency, used for severity prediction using images.

Supervised Learning: A machine learning approach where models are trained using labeled data, meaning the input data is paired with known output labels.

Demographic Data: Patient-related information such as age, gender, and body measurements that may influence disease risk.

Clinical Data: Medical information collected during patient examination, including symptoms, medical history, and physical assessments.

Biomarkers: Measurable biological indicators (e.g., blood test values) used to assess health conditions or disease presence.

SMOTE (Synthetic Minority Over-sampling Technique): A method used to balance imbalanced datasets by generating synthetic samples for the minority class.

Overfitting: A condition where a model performs well on training data but poorly on unseen data due to memorization rather than learning.

Underfitting: A condition where a model fails to capture underlying patterns in the data, resulting in poor performance on both training and testing data.

Generalization: The ability of a machine learning model to perform well on new, unseen data.

Fusion Model (Multi-modal Fusion) : A process by which outputs of multiple models are combined to generate a more precise and reliable prediction.

IoT (Internet of Things) : A network of interconnected devices that gather and exchange information in real time.

VAG (Vibroarthrography) : An analysis technique for measuring vibrations in joints for analyzing knee condition.

TinyML : A machine learning model that is able to run on microcontrollers and other small devices.

ESP32 : A type of microcontroller used in IoT solutions for collecting and processing data.

RMS (Root Mean Square) : Statistical measurement used in signal processing to estimate vibration signal magnitude.

Entropy (Signal Entropy) : A statistical metric used to measure randomness and complexity in signals.

Explainable AI (XAI) : Techniques for making machine learning models interpretable.

Grad-CAM (Gradient-weighted Class Activation Mapping) : A visualization technique that highlights important parts of an image in making a prediction.

API (Application Programming Interface): A system that defines methods of interaction between different software systems.

Accuracy: The proportion of correct predictions made by the model out of all predictions.

Precision: The ratio of correctly predicted positive observations to the total predicted positives.

Recall (Sensitivity): The ratio of correctly predicted positive observations to all actual positive cases.

F1-Score: The harmonic mean of precision and recall, providing a balance between the two.

Random Forest (RF): An ensemble learning method that builds multiple decision trees and combines their outputs for improved accuracy.

Gradient Boosting (GB): An ensemble technique that builds models sequentially, where each new model corrects errors made by previous ones.

XGBoost (Extreme Gradient Boosting): An optimized implementation of gradient boosting designed for efficiency and performance.

Hyperparameter Tuning: The process of optimizing model parameters to achieve the best performance.

Clinical Decision Support System (CDSS): A system designed to assist healthcare professionals in making clinical decisions using data-driven insights.

Local Interpretable Model-Agnostic Explanations (LIME) : Explanation of the prediction made by machine learning models for a specific observation.

SHapley Additive exPlanations (SHAP) : An explanation of the predictions made by model by allocating importance scores for each input feature.

Empirical Mode Decomposition (EMD) : Empirical Mode Decomposition is a method used to break down complex signals into smaller, simpler parts called Intrinsic Mode Functions.

Ensemble Empirical Mode Decomposition (EEMD) : An advanced form of EMD that adds small random noise to the signal and averages the results. This helps avoid mixing of frequencies and gives cleaner, more accurate signal components.”

Front End : User interface of the system which consists of user inputs/outputs.

Back End: Server side of the system which includes processing, modeling, etc.